1) Develop a Shopping cart in PHP with following requirements:

Admin Site:

1) Login

2) Manage Category (CRUD)

3) Manage Products (Product data with pictures. Upload picture and resize them)

4) Logout

User Site:

1) List Categories

2) Display products of category whose hyperlink is clicked(Apply pagination, load data with ajax).

3) Register user (Use CAPTCHA)

4) Login

Code:

------ shopping_cart/cart.php ------

```
<?php
require "db.php";
if (!isset($_SESSION['cart'])) {
   $_SESSION['cart'] = [];
}
if (isset($_GET['add'])) {
   $id = (int)$_GET['add'];
   $stmt = mysqli_prepare($conn, "SELECT id FROM products WHERE id=?");
   mysqli_stmt_bind_param($stmt, "i", $id);
   mysqli_stmt_execute($stmt);
   if (mysqli_fetch_assoc(mysqli_stmt_get_result($stmt))) {
      $_SESSION['cart'][$id] = ($_SESSION['cart'][$id] ?? 0) + 1;
   }
   header("Location: cart.php");
```

```
    exit;
}
if (isset($_GET['remove'])) {
    unset($_SESSION['cart'][(int)$_GET['remove']]);
    header("Location: cart.php");
    exit;
}
if (isset($_POST['clear'])) {
    $_SESSION['cart'] = [];
    header("Location: cart.php");
    exit;
}
$total = 0;
?>
<!doctype html>
<html><head><meta charset="utf-8"><title>My Cart</title><link rel="stylesheet"
href="css/style.css"></head>
<body>
<header><h1>My Shopping Cart</h1><nav><a href="index.php">Home</a><a
href="register.php">Register</a><a href="login.php">Login</a></nav></header>
<main class="container"><div class="panel">
<?php if (empty($_SESSION['cart'])): ?>
<p>Your cart is empty.</p>
<?php else: ?>
<table><tr><th>Product</th><th>Price</th><th>Quantity</th><th>Subtotal</th><th>Action</th></tr>
<?php foreach ($_SESSION['cart'] as $id => $qty):
    $id = (int)$id;
    $stmt = mysqli_prepare($conn, "SELECT name, price FROM products WHERE id=?");
    mysqli_stmt_bind_param($stmt, "i", $id);
    mysqli_stmt_execute($stmt);
    $product = mysqli_fetch_assoc(mysqli_stmt_get_result($stmt));
```

```
  if (!$product) continue;
  $subtotal = (float)$product['price'] * (int)$qty;
  $total += $subtotal;
?>
<tr>
<td><?= htmlspecialchars($product['name']) ?></td>
<td>₹<?= number_format((float)$product['price'], 2) ?></td>
<td><?= (int)$qty ?></td>
<td>₹<?= number_format($subtotal, 2) ?></td>
<td><a class="button danger" href="cart.php?remove=<?= $id ?>">Remove</a></td>
</tr>
<?php endforeach; ?>
</table>
<h3>Total: ₹<?= number_format($total, 2) ?></h3>
<form method="post"><button name="clear" value="1">Clear Cart</button></form>
<?php endif; ?>
<a class="button" href="index.php">Continue Shopping</a>
</div></main></body></html>
```

``````

------ shopping_cart/db.php ------

``````

```
<?php
if (session_status() === PHP_SESSION_NONE) {
  session_start();
}

$conn = mysqli_connect("localhost", "root", "krish123", "shopping_cart");

if (!$conn) {
```

```
  die("Database connection failed: " . mysqli_connect_error());
}
?>
```

``````

------ shopping_cart/index.php ------

``````

```
<?php
require "db.php";
$categories = mysqli_query($conn, "SELECT * FROM categories ORDER BY name");
?>
<!doctype html>
<html>
<head>
  <meta charset="utf-8">
  <title>Simple Shopping Cart</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
<header>
  <h1>Simple Shopping Cart</h1>
  <nav>
    <a href="index.php">Home</a>
    <a href="register.php">Register</a>
    <a href="login.php">User Login</a>
    <a href="cart.php">My Cart (<?= array_sum($_SESSION['cart'] ?? []) ?>)</a>
    <?php if (isset($_SESSION['user_id'])): ?>
      <a href="logout.php">Logout</a>
    <?php endif; ?>
    <a href="admin/login.php">Admin</a>
```

```
  </nav>
</header>
<main class="container">
  <h2>Shop by Category</h2>
  <div class="grid">
    <?php while ($cat = mysqli_fetch_assoc($categories)): ?>
      <div class="card">
        <h3><?= htmlspecialchars($cat['name']) ?></h3>
        <p class="muted">View products in this category.</p>
        <a class="button" href="products.php?category=<?= (int)$cat['id'] ?>">View Products</a>
      </div>
    <?php endwhile; ?>
  </div>
</main>
<footer>PHP and MySQL Shopping Cart Practical</footer>
</body>
</html>
```

``````

------ shopping_cart/login.php ------

``````

```
<?php
require "db.php";
$error = "";
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
  $email = trim($_POST['email'] ?? '');
  $password = $_POST['password'] ?? '';
  $stmt = mysqli_prepare($conn, "SELECT id, name, password FROM users WHERE email=?");
  mysqli_stmt_bind_param($stmt, "s", $email);
  mysqli_stmt_execute($stmt);
```

```
  $user = mysqli_fetch_assoc(mysqli_stmt_get_result($stmt));

  if ($user && password_verify($password, $user['password'])) {
    session_regenerate_id(true);
    $_SESSION['user_id'] = (int)$user['id'];
    $_SESSION['user_name'] = $user['name'];
    header("Location: index.php");
    exit;
  }
  $error = "Invalid email or password.";
}
?>
<!doctype html>
<html><head><meta charset="utf-8"><title>User Login</title><link rel="stylesheet"
href="css/style.css"></head>
<body>
<header><h1>User Login</h1><nav><a href="index.php">Home</a><a
href="register.php">Register</a></nav></header>
<main class="container"><div class="panel">
<h2>Login</h2>
<?php if ($error): ?><p class="error"><?= htmlspecialchars($error) ?></p><?php endif; ?>
<form method="post">
<label>Email</label><input type="email" name="email" required>
<label>Password</label><input type="password" name="password" required>
<button type="submit">Login</button>
</form></div></main></body></html>
```

``````

------ shopping_cart/logout.php ------

``````

```
<?php
```

```
require "db.php";

unset($_SESSION['user_id'], $_SESSION['user_name']);

header("Location: index.php");

exit;

?>
```

``````

------ shopping_cart/products.php ------

``````

```
<?php

require "db.php";

$category = max(0, (int)($_GET['category'] ?? 0));


if (isset($_GET['ajax'])) {

  header("Content-Type: text/html; charset=UTF-8");

  $page = max(1, (int)($_GET['page'] ?? 1));

  $limit = 4;

  $offset = ($page - 1) * $limit;


  $stmt = mysqli_prepare($conn, "SELECT COUNT(*) AS total FROM products WHERE category_id=?");

  mysqli_stmt_bind_param($stmt, "i", $category);

  mysqli_stmt_execute($stmt);

  $total = (int)mysqli_fetch_assoc(mysqli_stmt_get_result($stmt))['total'];


  $stmt = mysqli_prepare($conn, "SELECT id, name, price, image FROM products WHERE category_id=?
ORDER BY id DESC LIMIT ? OFFSET ?");

  mysqli_stmt_bind_param($stmt, "iii", $category, $limit, $offset);

  mysqli_stmt_execute($stmt);

  $result = mysqli_stmt_get_result($stmt);


  if (mysqli_num_rows($result) === 0) {
```

```
        echo "<p>No products found in this category.</p>";
    }

    while ($p = mysqli_fetch_assoc($result)) {
        echo '<div class="card product">';
        if ($p['image'] !== '') {
            echo '<img src="uploads/' . rawurlencode($p['image']) . '" alt="Product">';
        } else {
            echo '<div class="muted">No picture available</div>';
        }
        echo '<h3>' . htmlspecialchars($p['name']) . '</h3>';
        echo '<p>Price: ₹' . number_format((float)$p['price'], 2) . '</p>';
        echo '<a class="button" href="cart.php?add=' . (int)$p['id'] . '">Add to Cart</a>';
        echo '</div>';
    }

    $pages = (int)ceil($total / $limit);
    if ($pages > 1) {
        echo '<div class="panel">';
        for ($i = 1; $i <= $pages; $i++) {
            echo '<button type="button" onclick="loadPage(' . $i . ')">' . $i . '</button>';
        }
        echo '</div>';
    }
    exit;
}

$stmt = mysqli_prepare($conn, "SELECT name FROM categories WHERE id=?");
mysqli_stmt_bind_param($stmt, "i", $category);
mysqli_stmt_execute($stmt);
```

```
$cat = mysqli_fetch_assoc(mysqli_stmt_get_result($stmt));

if (!$cat) {

  http_response_code(404);

  $catName = "Category not found";

} else {

  $catName = $cat['name'];

}

?>

<!doctype html>

<html>

<head>

  <meta charset="utf-8">

  <title>Products</title>

  <link rel="stylesheet" href="css/style.css">

</head>

<body>

<header>

  <h1><?= htmlspecialchars($catName) ?></h1>

  <nav><a href="index.php">Home</a><a href="cart.php">My Cart</a><a href="login.php">User
Login</a></nav>

</header>

<main class="container">

  <h2>Products</h2>

  <div id="product-list" class="grid">Loading products...</div>

  <p><a class="button" href="index.php">Back to Categories</a></p>

</main>

<script>

function loadPage(page) {

  fetch('products.php?ajax=1&category=<?= $category ?>&page=' + page)

    .then(response => {

      if (!response.ok) throw new Error('Could not load products');
```

```
        return response.text();
      })
      .then(html => document.getElementById('product-list').innerHTML = html)
      .catch(() => document.getElementById('product-list').textContent = 'Unable to load products.');
}
loadPage(1);
</script>
</body>
</html>
```

``````

------ shopping_cart/register.php ------

``````

```
<?php
require "db.php";

if (!isset($_SESSION['captcha_a'], $_SESSION['captcha_b'])) {
   $_SESSION['captcha_a'] = random_int(1, 9);
   $_SESSION['captcha_b'] = random_int(1, 9);
}
$error = "";
$success = "";

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
   $name = trim($_POST['name'] ?? '');
   $email = trim($_POST['email'] ?? '');
   $password = $_POST['password'] ?? '';
   $answer = (int)($_POST['captcha'] ?? -1);

   if ($answer !== $_SESSION['captcha_a'] + $_SESSION['captcha_b']) {
```

```
    $error = "Incorrect CAPTCHA. Try again.";
    unset($_SESSION['captcha_a'], $_SESSION['captcha_b']);
  } elseif ($name === '' || !filter_var($email, FILTER_VALIDATE_EMAIL) || strlen($password) < 4) {
    $error = "Enter a valid name/email and a password of at least 4 characters.";
  } else {
    $hash = password_hash($password, PASSWORD_DEFAULT);
    $stmt = mysqli_prepare($conn, "INSERT INTO users (name, email, password) VALUES (?, ?, ?)");
    mysqli_stmt_bind_param($stmt, "sss", $name, $email, $hash);
    if (mysqli_stmt_execute($stmt)) {
      $success = "Registration successful. You can now log in.";
      unset($_SESSION['captcha_a'], $_SESSION['captcha_b']);
    } else {
      $error = "Registration failed. This email may already be registered.";
    }
  }
}
if (!isset($_SESSION['captcha_a'], $_SESSION['captcha_b'])) {
  $_SESSION['captcha_a'] = random_int(1, 9);
  $_SESSION['captcha_b'] = random_int(1, 9);
}
?>
<!doctype html>
<html><head><meta charset="utf-8"><title>Register</title><link rel="stylesheet"
href="css/style.css"></head>
<body>
<header><h1>User Registration</h1><nav><a href="index.php">Home</a><a
href="login.php">Login</a></nav></header>
<main class="container"><div class="panel">
<h2>Create Account</h2>
<?php if ($error): ?><p class="error"><?= htmlspecialchars($error) ?></p><?php endif; ?>
<?php if ($success): ?><p class="message"><?= htmlspecialchars($success) ?></p><?php endif; ?>
```

<form method="post">

<label>Name</label><input name="name" required>

<label>Email</label><input type="email" name="email" required>

<label>Password (minimum 4 characters)</label><input type="password" name="password" minlength="4" required>

<label>CAPTCHA: <?= (int)$_SESSION['captcha_a'] ?> + <?= (int)$_SESSION['captcha_b'] ?> = ?</label>

<input type="number" name="captcha" required>

<button type="submit">Register</button>

</form> </main></body></html>

``````

------ shopping_cart/setup.sql ------

``````

```
CREATE DATABASE IF NOT EXISTS shopping_cart;
USE shopping_cart;

CREATE TABLE IF NOT EXISTS categories (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL
);

CREATE TABLE IF NOT EXISTS products (
  id INT AUTO_INCREMENT PRIMARY KEY,
  category_id INT NOT NULL,
  name VARCHAR(100) NOT NULL,
  price DECIMAL(10,2) NOT NULL,
  image VARCHAR(255) DEFAULT '',
  FOREIGN KEY (category_id) REFERENCES categories(id)
    ON DELETE CASCADE
);
```

```
CREATE TABLE IF NOT EXISTS users (
   id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(100) NOT NULL,
   email VARCHAR(150) NOT NULL UNIQUE,
   password VARCHAR(255) NOT NULL
);

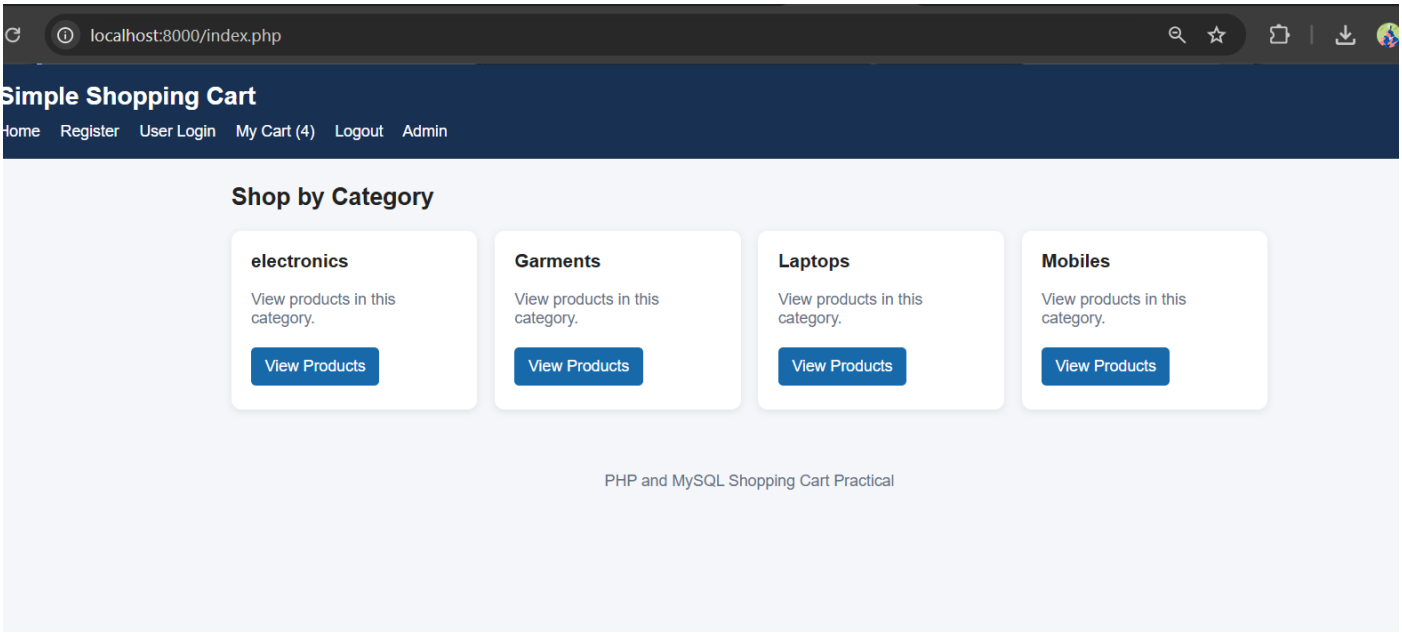

INSERT INTO categories (name)
SELECT 'Garments' WHERE NOT EXISTS
(SELECT 1 FROM categories WHERE name='Garments');

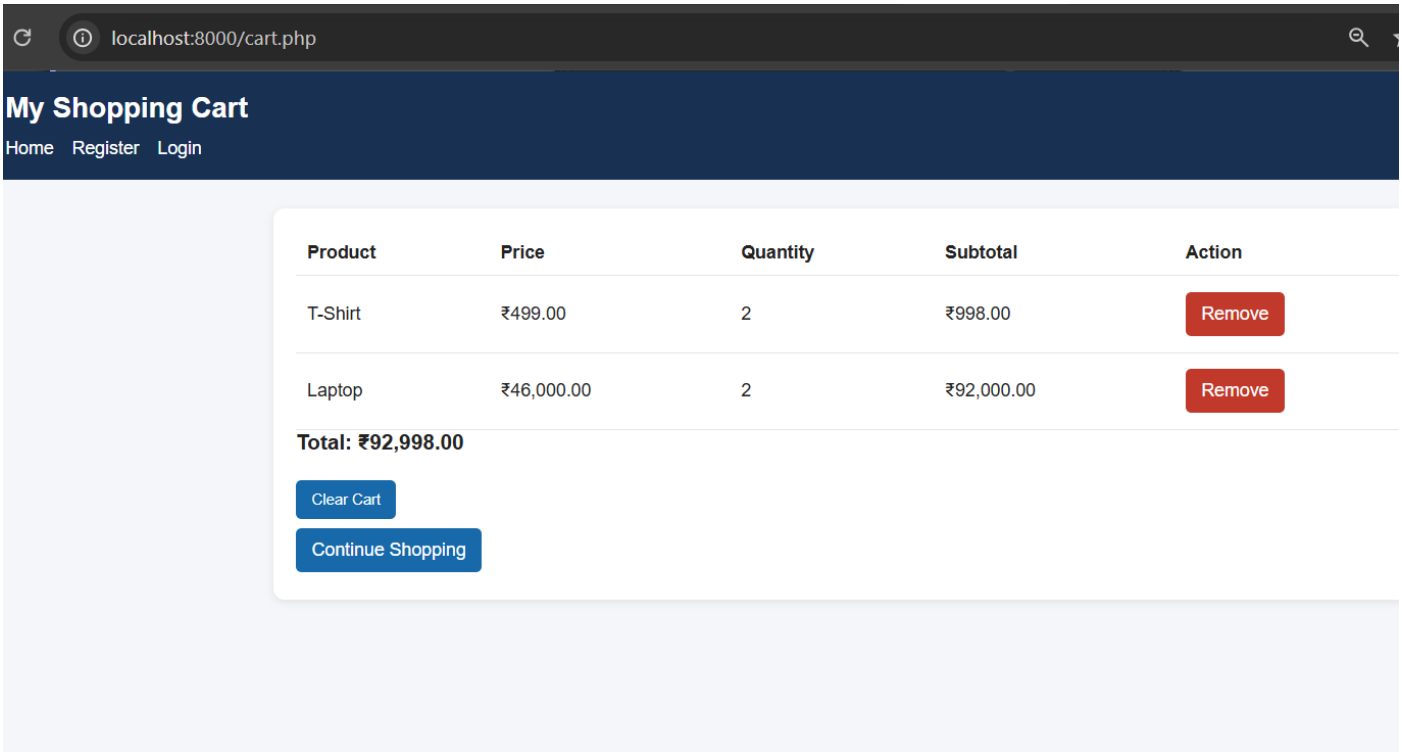

INSERT INTO categories (name)
SELECT 'Mobiles' WHERE NOT EXISTS
(SELECT 1 FROM categories WHERE name='Mobiles');


INSERT INTO categories (name)
SELECT 'Laptops' WHERE NOT EXISTS
(SELECT 1 FROM categories WHERE name='Laptops');


``````
```
```

output


localhost:8000/index.php
Simple Shopping Cart
Home Register User Login My Cart (4) Logout Admin
Shop by Category
electronics
View products in this category.
View Products
Garments
View products in this category.
View Products
Laptops
View products in this category.
View Products
Mobiles
View products in this category.
View Products
PHP and MySQL Shopping Cart Practical


localhost:8000/cart.php
My Shopping Cart
Home Register Login
Product Price Quantity Subtotal Action
T-Shirt ₹499.00 2 ₹998.00 Remove
Laptop ₹46,000.00 2 ₹92,000.00 Remove
Total: ₹92,998.00
Clear Cart
Continue Shopping

localhost:8000/admin/products.php

67% Reset

lanage Products

ashboard Categories Logout

Add Product

Category

Choose category

Product name

Price (₹)

Product picture (JPG or PNG, max 2 MB)

Choose file No file chosen

Pictures are resized to fit within 300 × 300 pixels.

Save Product

Product List

Picture	Product	Category	Price	Actions
	t-shirt	Garments	₹499.00	Edit | Delete
	Laptop	Laptops	₹46,000.00	Edit | Delete
	Mobile Phone	Mobiles	₹9,999.00	Edit | Delete

localhost:8000/index.php

hopping Cart

er User Login My Cart (4) Logout Admin

Shop by Category

electronics

View products in this category.

View Products

Garments

View products in this category.

View Products

Laptops

View products in this category.

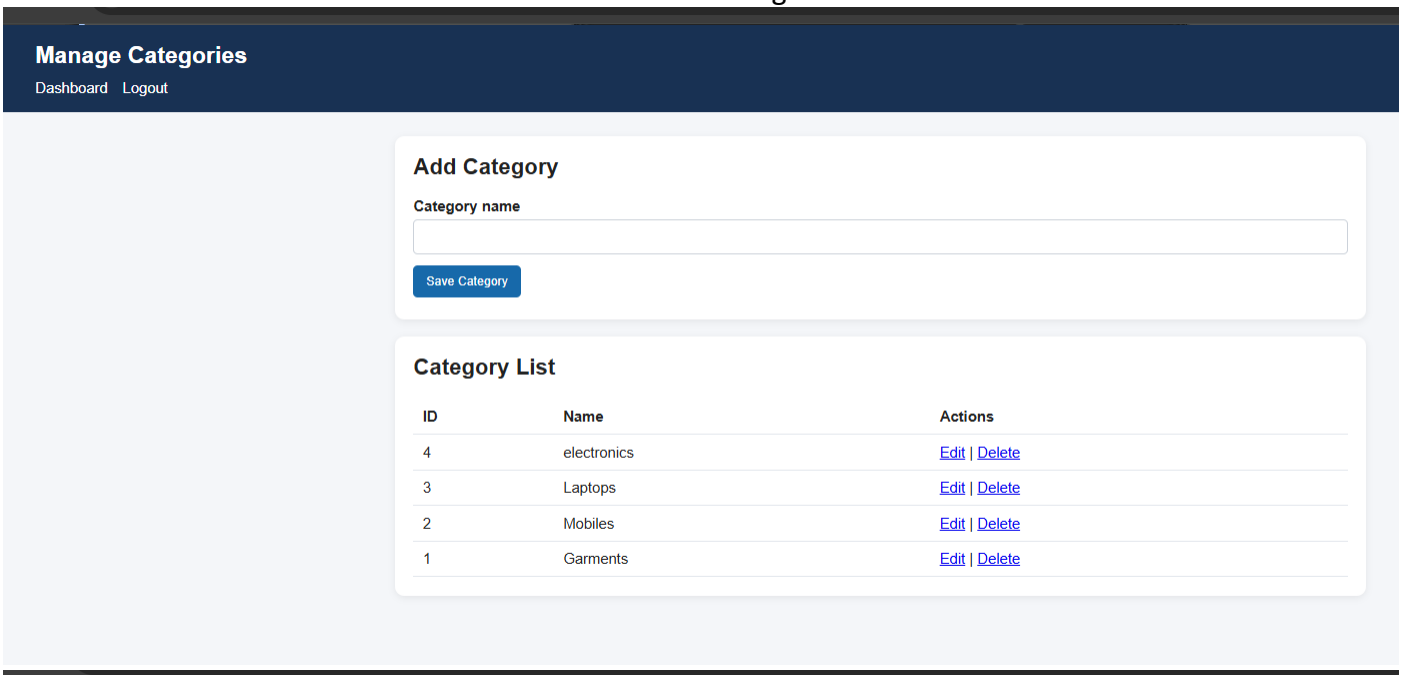
View Products

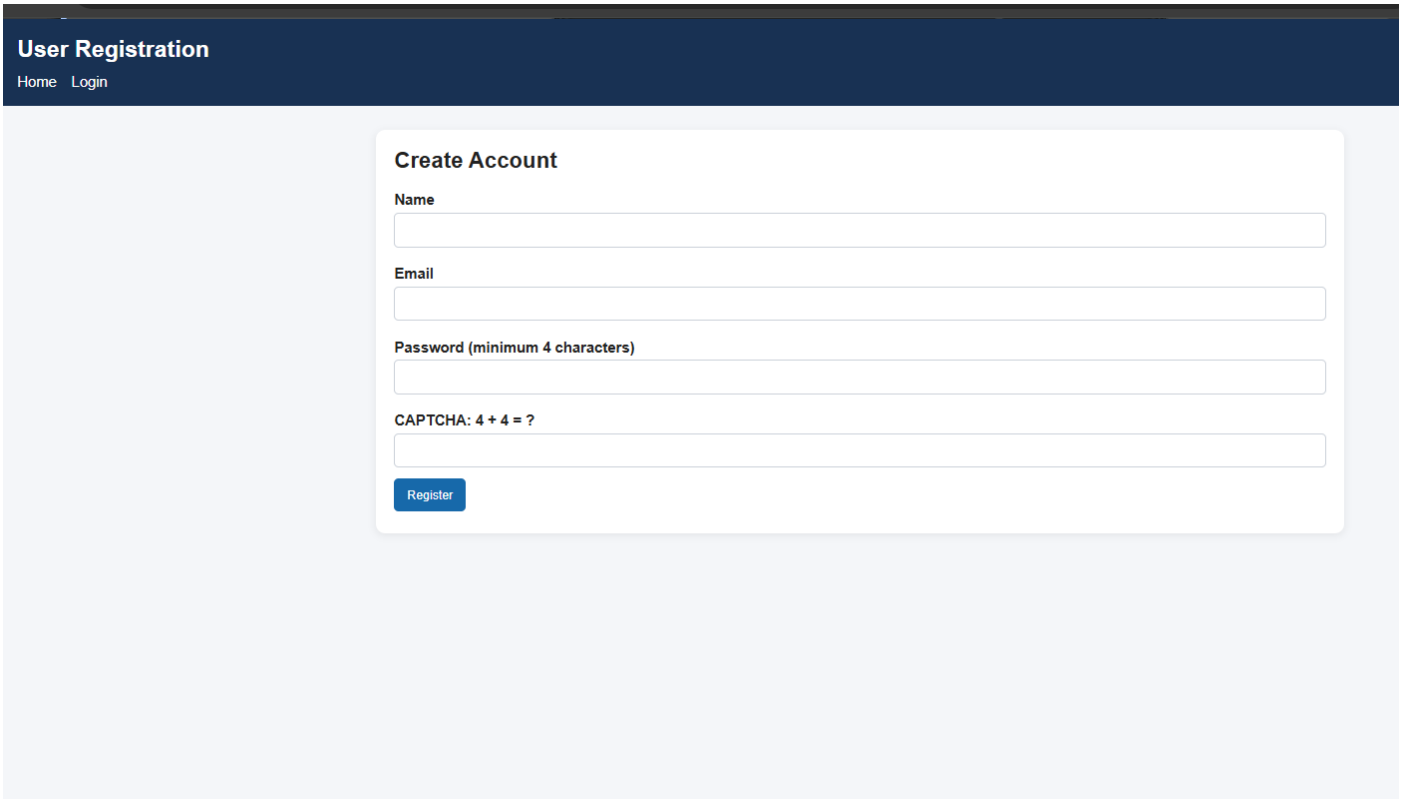
Mobiles

View products in this category.

View Products

PHP and MySQL Shopping Cart Practical


Manage Categories
Dashboard Logout
Add Category
Category name
Save Category
Category List
ID
Name
Actions
4
electronics
Edit | Delete
3
Laptops
Edit | Delete
2
Mobiles
Edit | Delete
1
Garments
Edit | Delete


User Registration
Home Login
Create Account
Name
Email
Password (minimum 4 characters)
CAPTCHA: 4 + 4 = ?
Register

2) Read data from students.xml file and store them in students table of MySQL.

Code:

Db.php

```
<?php
$conn = mysqli_connect(
   "localhost", "root", "krish123", "studentdb"
);

if (!$conn) {
   die("Connection failed");
}
?>
```

import.php

```
<?php
include "db.php";

$xml = simplexml_load_file("students.xml");

if ($xml === false) {
   die("XML file not found or invalid");
}

foreach ($xml->student as $s) {
   $id = (int)$s->id;
   $name = mysqli_real_escape_string($conn, (string)$s->name);
   $course = mysqli_real_escape_string($conn, (string)$s->course);
```

```
    $sql = "INSERT INTO students (id, name, course)
            VALUES ($id, '$name', '$course')";

    if (mysqli_query($conn, $sql)) {
        echo "Student inserted: " .
            htmlspecialchars($name) . "<br>";
    } else {
        echo "Insert failed for ID $id: " .
            mysqli_error($conn) . "<br>";
    }
}

mysqli_close($conn);
?>
```

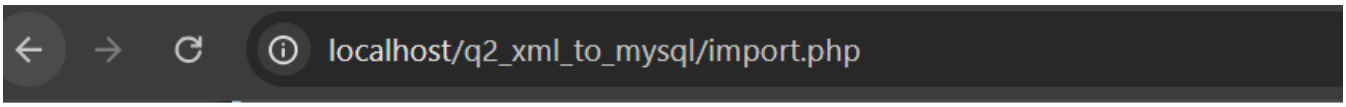
students.xml

```
<?xml version="1.0"?>
<students>
    <student>
        <id>1</id>
        <name>Rahul</name>
        <course>MSc ICT</course>
    </student>

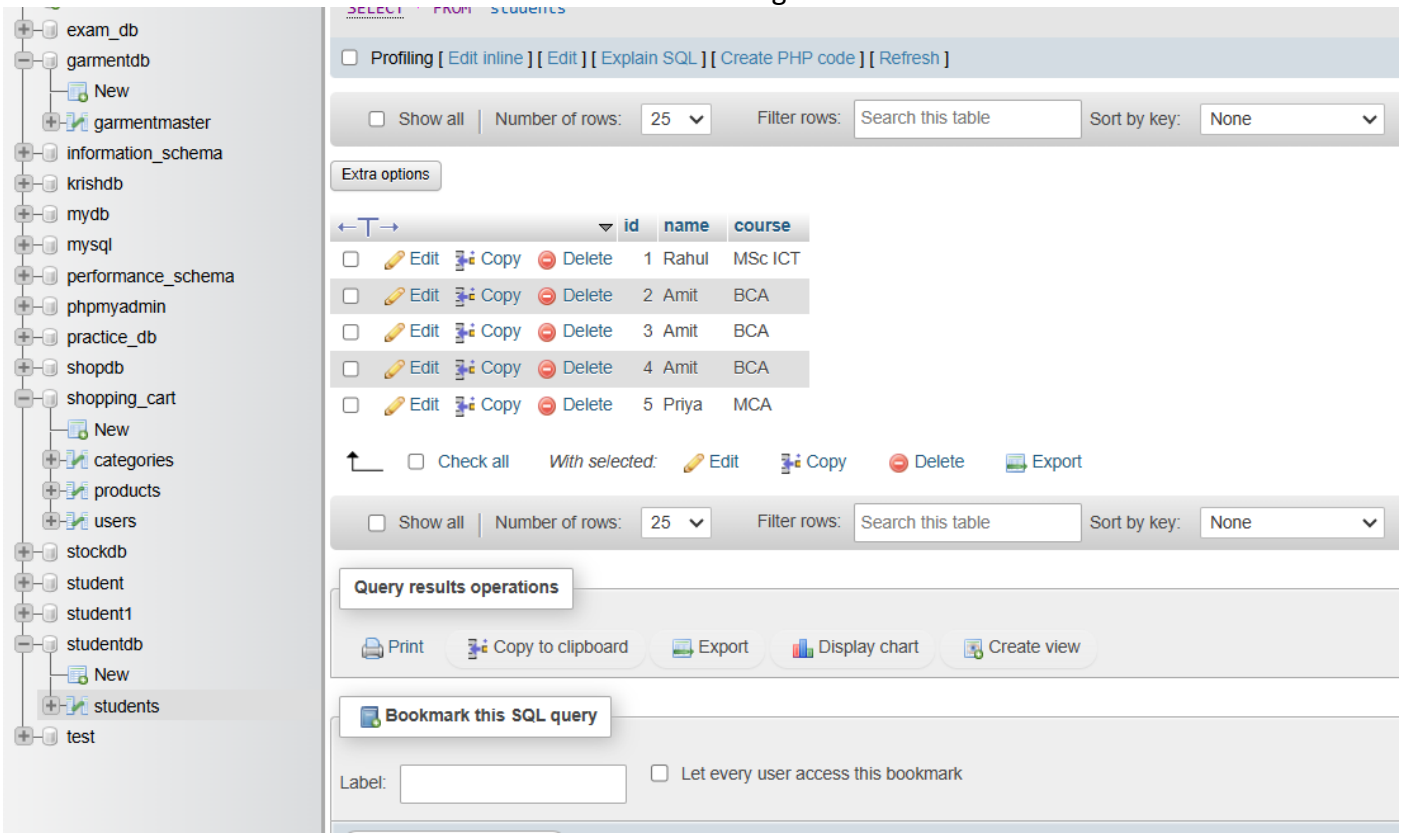
    <student>
        <id>2</id>
        <name>Amit</name>
        <course>BCA</course>
    </student>
```

```
  <student>
    <id>3</id>
    <name>Amit</name>
    <course>BCA</course>
  </student>
  <student>
    <id>4</id>
    <name>Amit</name>
    <course>BCA</course>
  </student>
  <student>
    <id>5</id>
    <name>Priya</name>
    <course>MCA</course>
  </student>
</students>
```

Output :



tudent inserted: Rahul
tudent inserted: Amit
tudent inserted: Amit
tudent inserted: Amit
tudent inserted: Priya



3) Read data from students table of MySQL and store them in students.xml file.

Code:

Db.php

```
<?php
$conn = mysqli_connect(
   "localhost", "root", "krish123", "studentdb"
);

if (!$conn) {
   die("Connection failed");
}
?>
```

exports.php

```
<?php
include "db.php";
```

```
$result = mysqli_query($conn, "SELECT * FROM students");

$xml = new SimpleXMLElement("<students/>");

while ($row = mysqli_fetch_assoc($result)) {
   $student = $xml->addChild("student");

   $student->addChild("id", $row['id']);
   $student->addChild("name", $row['name']);
   $student->addChild("course", $row['course']);
}

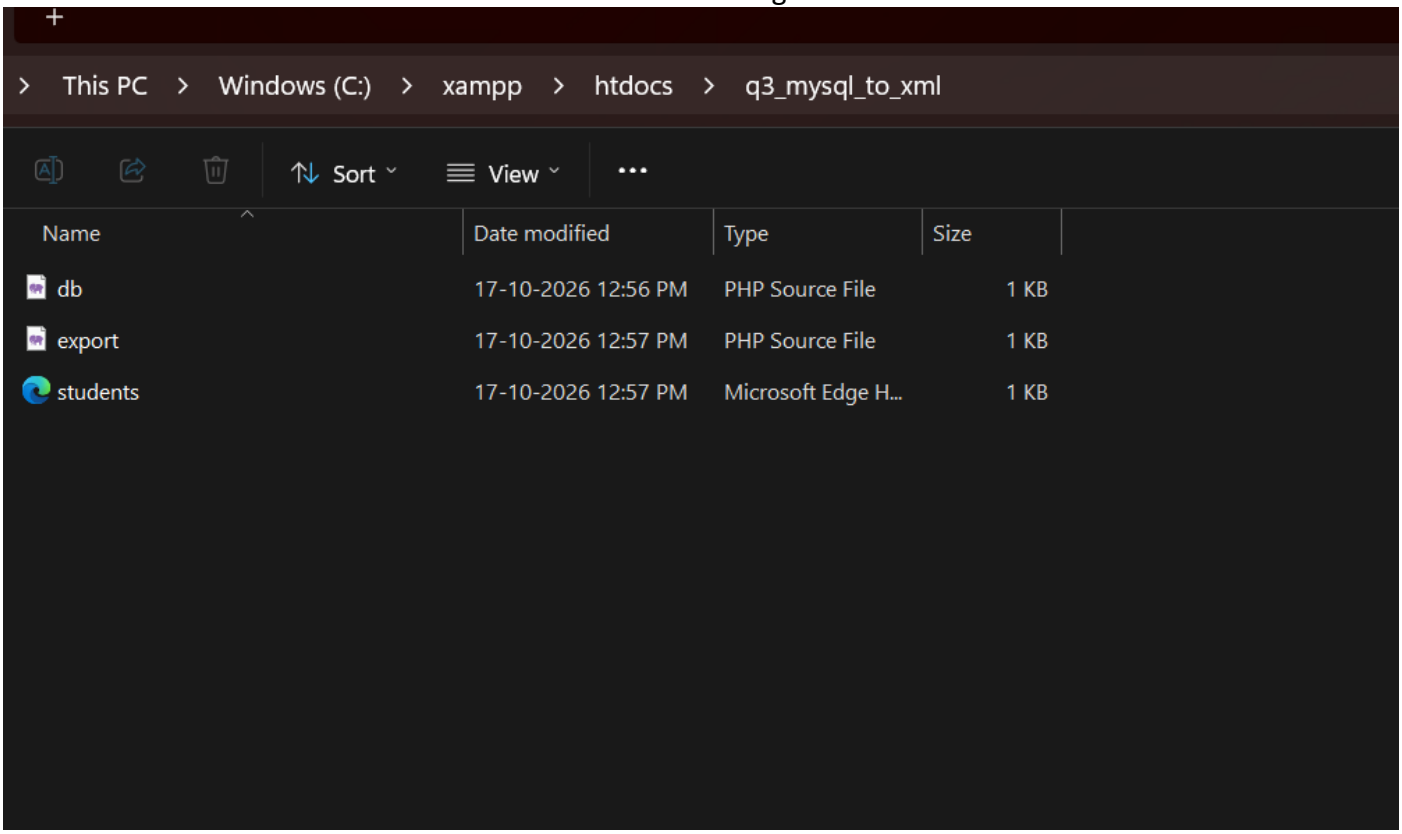
$xml->asXML("students.xml");

echo "Students exported to XML successfully!";
?>
```

Output :



Students exported to XML successfully!



Students.xml :

<?xml version="1.0"?>

<students><student><id>1</id><name>Rahul</name><course>MSc ICT</course></student><student><id>2</id><name>Amit</name><course>BCA</course></student><student><id>3</id><name>Amit</name><course>BCA</course></student><student><id>4</id><name>Amit</name><course>BCA</course></student><student><id>5</id><name>Priya</name><course>MCA</course></student></students>

4) List files in a directory.

Code:

Index.php

```
<?php
$folder = "myfiles";

if (is_dir($folder)) {
```

```
    $files = scandir($folder);


    echo "<h2>Files in Directory</h2>";


    foreach ($files as $file) {
        if ($file != "." && $file != "..") {
            echo $file . "<br>";
        }
    }
} else {
    echo "Directory not found";
}
?>
```

Output :



Files in Directory

demo.txt
notes.txt

5) Call express rest api in php.

Code:

Index.php

```
<?php
$url = "http://localhost:3000/students";

$response = file_get_contents($url);

if ($response === false) {
   die("API request failed");
}

$students = json_decode($response, true);

echo "<h2>Student List</h2>";

foreach ($students as $student) {
   echo "ID: " . $student['id'] . "<br>";
   echo "Name: " . htmlspecialchars($student['name']) . "<br>";
   echo "Course: " . htmlspecialchars($student['course']) . "<br><br>";
}
?>
```

Server.js

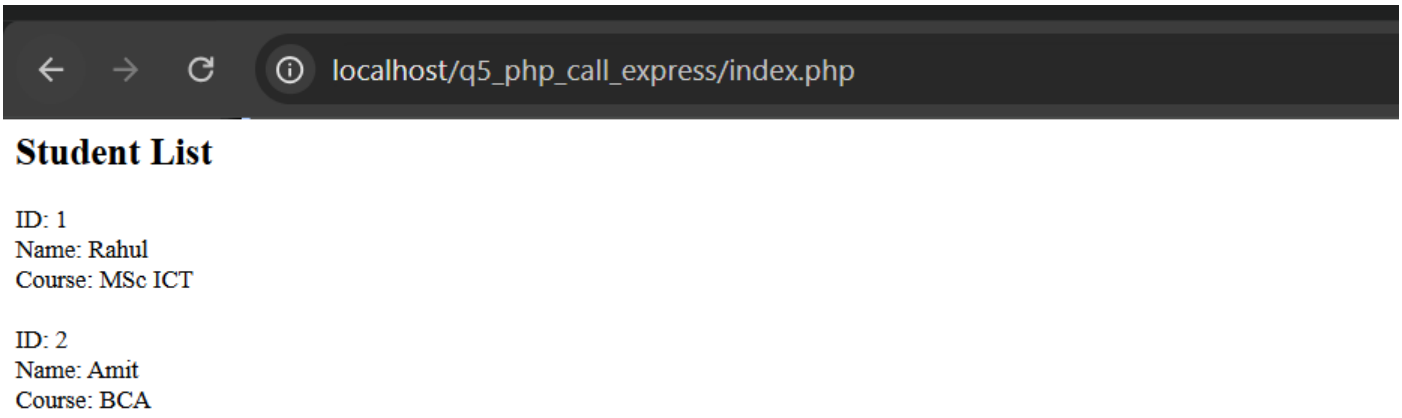
```
const express = require('express');

const app = express();
```

```
app.get('/students', (req, res) => {
   res.json([
      { id: 1, name: 'Rahul', course: 'MSc ICT' },
      { id: 2, name: 'Amit', course: 'BCA' }
   ]);
});

app.listen(3000, () => {
   console.log('Server running on port 3000');
});
```

Output :



6) Call php rest api in express.

Code:

Students.php

```
<?php
header("Content-Type: application/json");

$students = [
   [
      "id" => 1,
      "name" => "Rahul",
      "course" => "MSc ICT"
   ],
   [
      "id" => 2,
      "name" => "Amit",
      "course" => "BCA"
   ]
];

echo json_encode($students);
?>
```

Client Side (Server.js) :

```
const express = require('express');

const app = express();

app.get('/students', async (req, res) => {
   try {
```

```
    const response = await fetch(
      'http://localhost/q6_php_api/students.php'
    );

    if (!response.ok) {
      return res.status(502).send('PHP API failed');
    }

    const students = await response.json();

    res.json(students);
  } catch (error) {
    res.status(500).send('Error calling PHP API');
  }
});

app.listen(3001, () => {
  console.log('Express server running on port 3001');
});
```

Output :

